

REVIEW

Beyond the Cell

The cell biology of the hepatocyte: A membrane trafficking machine

key voxels for liver

membrane and lumen, cytoplasm [hɪ'pætəsaɪt]:hepatocyte

As in most cells, the origin of proteins destined for secretion from the hepatocyte begins with ribosomal synthesis and transport of nascent polypeptides through the Sec61 translocon into the lumen of the ER ahead of vesicular trafficking to the Golgi apparatus ([Rapoport et al., 2017](#)). It is within the ER and Golgi lumina that chaperone-assisted folding and (in the case of most secretory proteins) glycosylation occur ([Braakman and Bulleid, 2011](#); [Moremen et al., 2012](#)). Proteins exported from the hepatocyte are packaged into Golgi-derived secretory vesicles for transport to and exocytosis at the basolateral membrane ([Saucan and Palade, 1994](#)).

Detoxification of the blood and alcoholic liver injury

By virtue of its extensive vascularization, everything we eat, drink, and breathe passes through the liver. This positions the organ to detoxify the multitude of endotoxins and exotoxins we encounter on a daily basis. In general, fat-soluble toxins enter the liver and are metabolized by over 50 different cytochrome P450 enzymes embedded in the ER membrane ([Zanger and](#)

1. "er_mem", "er_lum", "er", "er_mem_all",
"golgi_mem", "golgi_lum", "golgi"

prior rules for heuristic method designing

1. In diseased hepatocytes, ER dilation and close spatial association between ER, lipid droplets, and mitochondria have been observed.

```
# disease conditions, pic A and B
low_cost_pairs = [
    ("er_mem", "ld_mem"),
    ("er_mem", "mito_mem"),

    ("er_mem", "ves_mem"),
    ("golgi_mem", "ves_mem"),
]
```

Lipoproteins, lipid droplets (LDs), lipolysis, and lipophagy

The liver is a critical depot for lipid uptake, storage, breakdown, and release. The **hepatocyte** takes up dietary lipid (in the form of chylomicron remnants) and **releases a substantial amount of lipid** back into the circulation as very low-density lipoprotein (**VLDL**) particles. These 30- to 80-nm structures are repositories

we only know part of the process as depicted in pic B

these pathways have been studied for nearly half a century, **the trafficking of VLDL particles from the ER and TGN and eventual secretion from the hepatocyte into the bloodstream remain largely undefined.** While there is evidence for the existence of

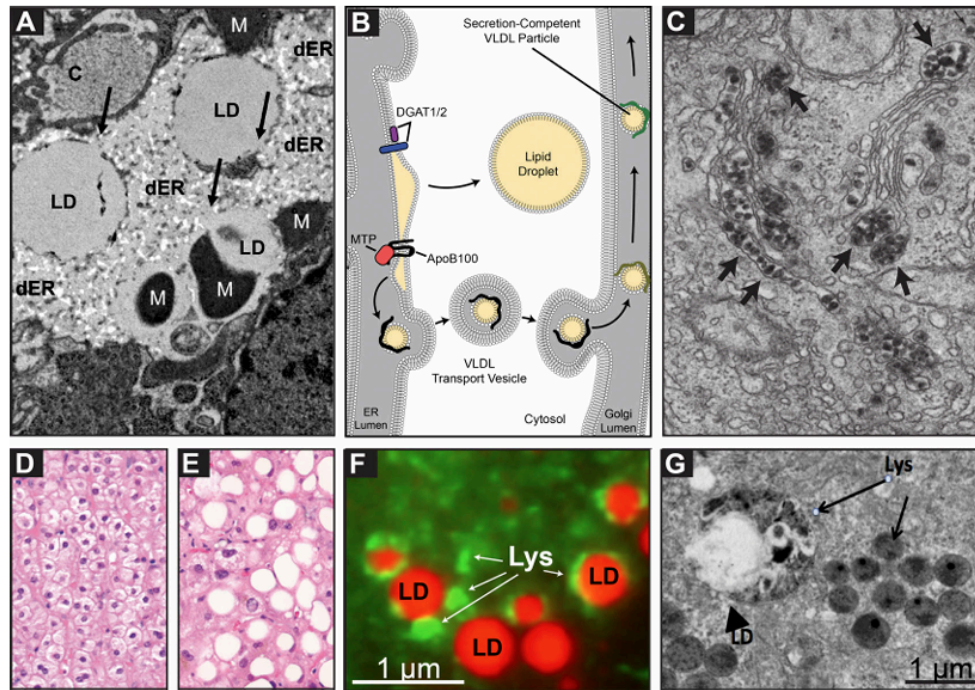


Figure 4. **Lipid accumulation and catabolism in hepatic disease.** (A) Ultrastructure of a "ballooned" human hepatocyte, a hallmark of lipid-induced inflammatory liver diseases such as nonalcoholic steatohepatitis, showing the dilated lumen of the ER (labeled dER) and its intimate connection with LDs, as indicated by arrows, and mitochondria (M). Panel A is reprinted with permission from Caldwell et al. (2010). (B) Cartoon illustrates the synthesis of cytosolic LDs and lipoprotein particles and trafficking stages between the ER, cytosol, and Golgi apparatus. DGAT, diacylglycerol acyltransferase; MTP, microsomal triglyceride transfer protein. (C) Diseased rat hepatocyte showing the accumulation of lipoprotein particles within the secretory Golgi compartment (arrows) following alcohol-induced liver damage. Fig. 4 C was modified with permission from Ehrenreich et al. (1973). Magnification of 50,000. (D and E) Histology of human liver tissue shows the morphology of normal hepatocytes (D) versus the dramatic accumulation of LDs within steatotic hepatocytes (E). At the cellular level, LDs are catabolized in part by lysosomal digestion via lipophagy as shown in F and G. The fluorescence micrograph (F) displays BODIPY-stained LDs in red that intimately associate with lysosomes stained positive by LAMP1 immunofluorescence in green. The electron micrograph (G) illustrates the ultrastructure of this interaction and close proximity of lysosomes to LDs in hepatocytes.

```
# pic B
high_cost_pairs = [
    ("ld_lum", "lyso_lum"),
    ("ld_lum", "cyto"),
    ("lyso_lum", "cyto"),
]
```

```
# pm = plasma membrane ecs = extracellular space
```

```
low_cost_pairs = [  
    ("cyto", "pm"),  
    ("pm", "ecs"),  
]
```

```
high_cost_pairs = [  
    ("cyto", "ecs"),  
]
```

"ecs", "pm", "mito_mem", "mito_lum", "mito_ribo", "golgi_mem", "golgi_lum", "ves_mem", "ves_lum",
"endo_mem", "endo_lum", "lyso_mem", "lyso_lum", "ld_mem", "ld_lum", "er_mem", "er_lum",
"eres_mem", "eres_lum", "ne_mem", "ne_lum", "np_out", "np_in", "hchrom", "echrom", "nucpl", "mt_out",
"cyto", "mt_in", "nuc", "golgi", "ves", "endo", "lyso", "ld", "eres", "perox_mem", "perox_lum", "perox", "mito",
"er", "ne", "np", "chrom", "mt", "cell", "er_mem_all", "ne_mem_all